	SRM Institute of Science and Technology **Department of Mathematics** **21MAB204T-Probability and Queueing Theory** **Unit – I** **Tutorial Sheet - 3**
	Part – B (Each question carries 8 marks)
1	An unbiased coin is tossed 100 times. Use Tchebycheff's inequality to find a lower bound for the probability of getting 30 to 70 fours.
2	A Random Variable X has PDF $f(x) = e^{-x}, x \geq 0$. Use Tchebycheff's inequality to find $P\ (\lvert X - 1\rvert > 1)$ and compare it with the actual probability.
3	If the Tchebycheff's inequality for a R. V X with SD 3 is $P\ (6 < X < 18) \geq \frac{3}{4}$ find the mean of X.
4	Use Tchebycheff's inequality to prove that in 1000 throws with a coin the probability that the number of heads lies between 450 and 550 is at least $9 / 10$
5(i)	If X is a RV uniformly distributed in the interval (1, 2) find the Probability Density Function of $Y = e^X$
(ii)	If X is a Random Variable uniformly distributed in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, find the PDF of $Y = tanX$.
	Part – C (15 marks)
6	If X denotes the sum of the numbers obtained when two dice are thrown, obtain an upper bound for $P(\lvert X - 7\rvert \geq 4)$. Compare it with the exact probability.
7	Use Tchebycheff's inequality to determine how many times a fair coin must be tossed in order that the probability will be at least 0.90 that the ratio of the observed number of heads to the number of tosses will lie between 0.4 and 0.6.
8(i)	If X is the number on a die when it is thrown, prove that P (|x-μ|>2.5) < 0.47, where μ is mean.
(ii)	Let X be a continuous RV whose probability density function given by f(x)= e^{-x} $0 \leq x \leq \infty$. Using Tchebycheff's inequality verify P(|X-μ|>2) $\leq \frac{1}{4}$ and show that actual probability is e^{-3}.